

## 5.4 Applications of Integration

**Theorem (Net Change Theorem):** The integral of a rate of change is the *net change*:

$$\int_a^b F'(x) dx = F(b) - F(a)$$

Examples:

- Integral of *volume/sec* is net change in *volume*.
- Integral of *tons of pollution/hour* is net change in *tons of pollution*.
- Integral of *velocity* is the net change in *position*.

**Example 1:** Suppose that oil starts leaking from a tank at a rate of  $100 - 5t$  gallons per minute at time  $t$  from when the leaking starts.

a) What does  $\int_0^{120} (100 - 5t) dt$  represent?

b) Find the total number of gallons leaked in the first half-hour.

**Example 2:** At 4:00am, the layer of ice on a lake was 6in. thick. Its thickness is increasing at a rate of  $0.2t$  inches per hour where  $t$  represents the number of hours since 4:00am.

a) How thick was the ice at noon that same day?

b) How much thickness was gained from 6:00am to noon that same day?

**Note:** Finding the *total distance* travelled is different than computing the *total displacement*.

- *Displacement* represents a *Net Change* and so quantities of distance are measured as positive or negative.
- With *total distance*, all quantities are regarded as positive.

**Example 3:**

A particle moves along a line so that its velocity at time  $t$  is  $v(t) = t^2 - t - 6$  (measured in meters per second).

- a) Find the displacement of the particle during the time period  $1 \leq t \leq 4$ .

- b) Find the distance traveled during this time period.